# TOPIC 2 Transforming and Comparing Linear Functions

Name _____ Date _____

**Reminders**

Slope = $(y_2 - y_1) \div (x_2 - x_1)$ · $y = mx + b$ · point-slope: $y - y_1 = m(x - x_1)$

Parallel: same slope · Perpendicular: flip the fraction and change the sign (product $-1$)

Outside f(x) + k: up/down · Inside f(x − h): right h, f(x + h): left h (opposite the sign)

Horizontal: y = #, slope 0 · Vertical: x = #, slope undefined (HOY VUX)

Domain = inputs (x) · Range = outputs (y) · Zero = x-intercept

**1.** Kai graphed $f(x)$ and $g(x) = f(x + 5)$. Which statement describes how the graphs are related?

**Tip:** Inside the parentheses: move the opposite way of the sign.

**A.** translated right 5 units

**B.** translated up 5 units

**C.** **translated left 5 units**

Inside the parentheses the graph moves opposite the sign: + 5 → left 5.

**2.** A pool rental costs $50 plus $30 per hour, for at most 4 hours. What is the range for one rental?

> **Tip:** Range = the y-values (dollars). Find the cost at 0 hours and at 4 hours.

**F.** **50 ≤ y ≤ 170**

**G.** 0 ≤ x ≤ 4

**H.** 30 ≤ y ≤ 170

Range = costs. 0 hours: $50. 4 hours: 50 + 30(4) = $170.

**3.** What is the slope of a line perpendicular to $y = -4/5x + 2$?

> **Tip:** Flip the fraction and change the sign.

**A.** −5/4

**B.** **5/4**

**C.** 4/5

Flip −4/5 to −5/4 and change the sign: 5/4. Check: (−4/5)(5/4) = −1.

**4.** A line passes through (0, −3) and (6, 0). What is the zero of the function?

> **Tip:** The zero is the x-value where the line crosses the x-axis.

**F.** −3

**G.** 1/2

**H.** 6

The zero is where y = 0: the point (6, 0). −3 is the y-intercept.